The Koide relation from the $\mathbb{Z}_3$ symmetry of the trefoil

A topological reading of $Q = 2/3$ ($r = \sqrt{2}$ *carried* by the sense doubling — conditional), Koide's (1981) relation for m_τ, the phase decomposed as $\delta = \pi/12 - \varepsilon$, and the closed rail-magnetism chain : m_e computed to $\pm 3\,\%$ (TQNT)

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Abstract

Koide's relation $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ is among the most precise numerical coincidences in particle physics (satisfied to 9.2 ppm), yet the Standard Model offers *no* explanation: the three lepton masses are three free Yukawa couplings. Within the Topological Knot Quantum Theory (TQNT), where the charged lepton is the trefoil knot $T(2,3) = 3_1$, we show that the $\mathbb{Z}_3$ **symmetry** of the trefoil enforces exactly the three-mode structure of the Koide parametrisation, and that the amplitude $r = \sqrt{2}$ — postulated in v1 of this paper — now has a *carrier* (conditional): the lepton doublet is the doublet of the **two circulation senses** of the rail current on the trefoil, and two channels against one give $r = d_{1/2}(k{=}2) = \sqrt{2}$ (the same $\sqrt{2}$ as the exchange excess $d - 1$ of the heavy sector of the monograph). One recovers **Koide's (1981) relation**, predicting the tau mass from m_e and m_μ: $m_\tau^{\mathrm{pred}} = 1776.97$ MeV against 1776.86 MeV measured (+61 ppm). The phase, an orphan hint in v1 ($\delta = 2/9$), is *decomposed*: $\delta = \pi/12 - \varepsilon$, where $\pi/12 = 2\pi/24$ is *one step* of the trefoil's minimal lattice circuit (24 steps, Diao's theorem) and $\varepsilon = 0.0396 \simeq \sqrt{m_e/\mu}$ measures the quasi-nodality of the electron — its lightness is a **nodal protection**. The chain closes with a computation: m_e = *retarded* Green energy of the framing-line regrouping mode of the ideal trefoil tube (n=3, $m = +1$ selected by line-current conservation) $\to$ 0.518–0.522 MeV, i.e. m_e to $+1.4/+2.2\,\%$, *zero parameters*. The scale itself is identified: $m_0 = \frac{3}{4}\sqrt{\sigma_\chi} = 313.85$ MeV against 313.84 extracted from the data ($\sim 5 \times 10^{-5}$). We separate explicitly what is derived, conditional, postulated and confirmed, and give an honest epistemic assessment.

1 The puzzle of the charged-lepton mass hierarchy

The charged-lepton masses (PDG 2024) [5]

$$m_e = 0.510\,998\,95 \text{ MeV}, \quad m_\mu = 105.658\,375\,5 \text{ MeV}, \quad m_\tau = 1776.86 \text{ MeV}$$

span four orders of magnitude with no apparent structure. In the Standard Model they are *three free parameters*. In 1981–83 Koide [1] noted that the dimensionless combination

$$Q \;\equiv\; \frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} \tag{1}$$

takes, on the measured values,

$$Q_{\mathrm{exp}} = 0.666\,660\,5 \;\approx\; \tfrac{2}{3} = 0.666\,666\,7 \qquad (\text{deviation } -9.2 \text{ ppm}).$$

Foot [2] gave it a geometric reading ($Q = 2/3$ is the angle 45° between the vector $(\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$ and $(1,1,1)$); Brannen [3] recast it with an explicit phase; Sumino [4] sought a family gauge origin. None of these is part of the Standard Model, which leaves the coincidence unexplained.

2 Framework: the lepton is the trefoil 3_1

In TQNT the charged lepton is the trefoil $T(2,3) = 3_1$ (the simplest nontrivial knot: crossing number cr = 3, braid index $b = 2$). Its decisive feature here is its $\mathbb{Z}_3$ **symmetry**: the torus knot $T(2,3)$ is invariant under a $2\pi/3$ rotation of its braid presentation. The **three generations** (e, μ, τ) are identified with the **three eigenmodes** of this $\mathbb{Z}_3$ symmetry, i.e. the three cube roots of unity $\omega^k = e^{2\pi ik/3}$, $k = 0, 1, 2$.

3 Derivation of the Koide form

The three-mode structure (DERIVED). A $\mathbb{Z}_3$ eigenmode has a real amplitude $\propto \cos(\delta + 2\pi k/3)$. Since the mass is the squared energy of the χ-field amplitude,

$$\sqrt{m_k} = \sqrt{m_0}\left(1 + r\cos\!\left(\delta + \tfrac{2\pi k}{3}\right)\right), \qquad k = 0, 1, 2. \tag{2}$$

The cube-root-of-unity identities $\sum_k \cos(\delta + 2\pi k/3) = 0$ and $\sum_k \cos^2(\delta + 2\pi k/3) = \frac{3}{2}$ give, *independently of* δ,

$$\sum_k \sqrt{m_k} = 3\sqrt{m_0}, \qquad \sum_k m_k = m_0\left(3 + \tfrac{3}{2}r^2\right).$$

The value $r = \sqrt{2}$ (from postulate to carrier — conditional). Substituting into (1),

$$Q = \frac{m_0(3 + \frac{3}{2}r^2)}{(3\sqrt{m_0})^2} = \frac{1 + \frac{1}{2}r^2}{3}.$$

The $\mathbb{Z}_3$ symmetry alone does not fix r. Version 1 of this paper **postulated** it, after the failure of the direct attempts (Chern–Simons/Jones and Ising quantum dimensions; a $\mathbb{Z}_3$ tight-binding of the trefoil χ-modes: r fixed only up to a free scale); the v1 *Caveat* remains exact: the naive argument "the $\sqrt{2}$ of the spin-$\frac{1}{2}$ loop measure" is **wrong** (the SU(2) ladder operator J_+ gives $\sqrt{2}$ for *spin 1* and 1 for spin-$\frac{1}{2}$). The **carrier** identified since (rail magnetism, 2026-07-27, section 6) is of a different nature: the lepton doublet is the doublet of the *two circulation senses* of the rail current ($\pm j$ exact, degeneracy theorem); a mode coupling to *both* senses has two channels against one, and the amplitude ratio is the quantum dimension $d_{1/2}(k{=}2) = \sqrt{2}$ — *the same* $\sqrt{2}$ as the exchange excess $d - 1$ of the heavy baryons of the monograph: two readings of *one* structure (the channel doubling of the $k{=}2$ ribbon). Status: **conditional** (declared identification amplitude = channels), no longer a bare postulate. One then has **exactly**

$$\boxed{r = \sqrt{2} \iff Q = \tfrac{1+1}{3} = \tfrac{2}{3}}, \tag{3}$$

and $Q = 2/3$ is **CONFIRMED** to 9.2 ppm.

4 Koide's (1981) relation for m_τ

Being a single equation among the three masses, $Q = 2/3$ **predicts m_τ from the measured m_e and m_μ** — this is **Koide's historical 1981 relation** (he already predicted $m_\tau \approx 1777$ MeV), here reinterpreted via the trefoil's $\mathbb{Z}_3$:

$$\sqrt{m_\tau} = 2\left(\sqrt{m_e} + \sqrt{m_\mu}\right) + \sqrt{3\,(m_e + m_\mu) + 12\sqrt{m_e\, m_\mu}}\,, \tag{4}$$

giving

$$\boxed{m_\tau^{\text{pred}} = 1776.97 \text{ MeV}} \quad \text{vs.} \quad m_\tau^{\text{PDG}} = 1776.86 \text{ MeV} \quad (+61 \text{ ppm}).$$

This is a *falsifiable* prediction that could have returned any value; it lands 61 ppm from the measurement.

5 The full spectrum: the scale identified and the phase decomposed

Parametrisation (2) still contains the scale m_0 and the phase δ. Version 1 proposed $m_0 = m_p/3$ and $\delta = 2/9$; the 2026-07 state replaces them with:

- **scale** $m_0 = \frac{3}{4}\sqrt{\sigma_\chi} = 313.85$ MeV ($\sqrt{\sigma_\chi} = 418.47$ MeV, the single string tension of the monograph, recalibrated 2026-07-18): value extracted from data $m_0 = (\frac{1}{3}\sum_k \sqrt{m_k})^2 = 313.84$ MeV, agreement $\sim 5 \times 10^{-5}$ (identification; the historical reading $m_p/3 = 312.76$, $+0.35\,\%$, is its approximate version — the lepton sector and the baryonic base derive from the *same* scale);
- **phase** $\delta = \pi/12 - \varepsilon$ (decomposed, 2026-07-27): $\pi/12 = 2\pi/24$ is *one step* of the trefoil's minimal lattice circuit (24 steps, Diao's theorem; the $\mathbb{Z}_3$ modes fall at steps $\{0, 8, 16\}$, the amplitude node at step 9 — integers); $\varepsilon = 0.0396 \simeq \sqrt{m_e/\mu}$ is the quasi-nodality of the electron (section 6). The numerical value remains $\delta = 0.2222$ ($= 2/9$ to $0.05\,\%$ of the extracted angle $\theta_\tau = 0.2223$): the rational "2/9" of v1 was not the structure — the structure is $\pi/12 - \varepsilon$.

With these topological values and $r = \sqrt{2}$, the parametrisation reproduces all **three** masses:

	m^{pred} (MeV)	m^{PDG} (MeV)	deviation
e	0.51	0.511	$-0.35\,\%$
μ	105.29	105.658	$-0.35\,\%$
τ	1770.75	1776.86	$-0.34\,\%$

Three masses across four orders of magnitude, from *one* scale ($m_p/3$) and *two* topological numbers ($r = \sqrt{2}$, $\delta = 2/9$), to $\sim 0.4\,\%$.

6 The rail-magnetism chain (2026-07-27): from degeneracy to m_e

Between v1 and this v2, the monograph consolidated a complete chain that gives each Koide ingredient its mechanism (conditional on declared identifications); we summarize its five links.

(i) The degeneracy theorem. The inductance kernel of the trefoil circuit is C_3-invariant: the classes $k = 1, 2$ are *exactly* degenerate (verified to 5×10^{-16}) — no parity-even energy can separate μ from τ. The *minimal* breaker is the **circulation sense** (odd under time reversal); the mirror exchanges $\mu \leftrightarrow \tau$: the CPT structure of the triplet is *structural*.

(ii) $r = \sqrt{2}$ at its carrier. The doublet = the two senses ($\pm j$ exact) $\rightarrow$ two channels against one $\rightarrow$ $r = d_{1/2}(k{=}2) = \sqrt{2}$ (previous section).

(iii) The amplitude node and nodal protection. The amplitude $1 + \sqrt{2}\cos(\delta + 2\pi k/3)$ vanishes at $3\pi/4$ *exactly* (the same $\sqrt{2}$); the electron is *quasi-nodal*: $\delta = \pi/12 - \varepsilon$ with $\varepsilon = 0.0396 \simeq \sqrt{m_e/\mu}$. **The lightness of the electron is a nodal protection** (minimal overlap with the condensate), not an accident. At the exact lattice lock ($\delta = \pi/12$, integers only): $m_e = 0$ exactly, m_τ to $-1.1\,\%$, m_μ to $+19\,\%$ — the discrete skeleton carries the hierarchy to $\sim 20\,\%$, and ε does the rest.

(iv) The mode selection. The current is carried by the framing lines of the tube (each at fixed current: no source); the excitations are *regrouping* waves of the lines, and the condition $k \perp$ lines reads $3 + m\,\mathrm{Tw} = 0$, i.e. $m^* = 3/(\mathrm{Wr} - \mathrm{Lk}) = 1.24$: the only allowed integer mode is $m = +1$ (*kinematic* selection, sign fixed by chirality).

(v) m_e computed. The energy of the mode ($n{=}3$, $m{=}+1$) on the ideal trefoil tube, with the *retarded* Green kernel ($\cos(k\Delta r)/\Delta r$: the mode is a wave, $\omega = 3 \cdot 2\pi c/L$), gives

$$m_e^{\text{calc}} = 0.518\text{–}0.522 \text{ MeV} \qquad (m_e^{\text{PDG}} = 0.511\ ;\ +1.4/+2.2\,\%),$$

zero parameters (declared error budget ±3 %: geometry, quadrature, frustration, static calibration). Cross-check: the complete framing-current energy equals $\alpha\, m_\ell$ to $-0.3\,\%$ — the electromagnetic fraction of the tube recovered from geometry alone.

The complete chain of the lepton sector then reads: rail time → scale ($\frac{3}{4}\sqrt{\sigma_\chi}$); sense → generation (T-odd); two channels → $\sqrt{2}$; node at $3\pi/4 =$ step $9/24$ of the lattice; $\delta = \pi/12 - \varepsilon$; line conservation → $m = +1$; retarded Green → m_e. *Every link is conditional on a declared identification — none is a fit.*

7 Comparison with existing approaches

The Koide relation has received several explanations, each addressing a different facet.

Foot's geometric reading [2]. $Q = 2/3$ is equivalent to the vector $(\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$ making a 45° angle with the democratic direction $(1, 1, 1)$. This is an exact geometric restatement, but it offers no dynamical origin for the 45°.

Brannen's parametrisation [3]. Writing $\sqrt{m_k}$ as a circulant with a phase δ, the data give $\delta \approx 2/9$ — the same value obtained here. The form is phenomenological: it identifies the angle but not its origin.

Sumino's family gauge model [4]. A spontaneously broken SU(3) family symmetry, with a scalar potential engineered so that its vacuum enforces $Q = 2/3$, and heavy family gauge bosons arranged to cancel the QED radiative corrections that would otherwise spoil the relation at the pole masses. This is a genuine dynamical mechanism, at the cost of new heavy fields and a tuned potential.

The present (TQNT) reading. The novelty here is the *origin* of the three-mode circulant structure: it is not imposed by a chosen scalar potential but is the $\mathbb{Z}_3$ symmetry of a concrete topological object, the trefoil $T(2, 3)$, with which the charged lepton is identified. This (i) explains why there are exactly *three* generations (three $\mathbb{Z}_3$ modes); (ii) yields the circulant form for free (cube-root identities); (iii) *carries* the amplitude $r = \sqrt{2}$ through the sense doubling (conditional; the "braid/spin $b = 2$" route remains wrong: $\sqrt{2}$ is the spin-1 coefficient, not the spin-$\frac{1}{2}$ one); and (iv) *decomposes* the phase into $\pi/12 - \varepsilon$ (the minimal-circuit step + the electron's quasi-nodality) and *computes* m_e to ±3 %. No new gauge bosons are required.

Two honest caveats. First, like all the approaches above, the present work uses the physical (pole) masses; the QED/QCD running that Sumino's mechanism was designed to control is *not* addressed here, so the exactness of $Q = 2/3$ at the measured masses is taken as input. Second, the chain of section 6 is *conditional*: every link rests on a declared identification (amplitude = channels; loop ↔ half-turn; vertex locking on the lattice, named not proved); the "eigenfrequency" route remains empty (assignment freedom), and pure holonomy for δ is *rejected* (no mechanism). The distinctive, falsifiable content remains the parameter-free $Q = 2/3$ and m_τ — to which the chain adds an independent computation of m_e at ±3 %.

8 Honest epistemic assessment

Status of each ingredient

- **DERIVED**: the Koide *form* ($\sum\sqrt{m} = 3\sqrt{m_0}$, $Q = (1+r^2/2)/3$) is an *exact* consequence of the $\mathbb{Z}_3$ three-mode structure (cube-root identities); the equivalence $r = \sqrt{2} \Leftrightarrow Q = 2/3$ is algebra.
- **POSTULATED**: generations = $\mathbb{Z}_3$ modes of the trefoil (the framework's bedrock). **Conditional** (v2, declared identifications): $r = \sqrt{2}$ (carried by the sense doubling); $\delta = \pi/12 - \varepsilon$ ($\pi/12$ derived from the minimal circuit, ε computed); $m_0 = \frac{3}{4}\sqrt{\sigma_\chi}$ (identification to 5×10^{-5}, $\sqrt{\sigma_\chi}$ calibrated in the monograph).
- **CONFIRMED**: $Q = 2/3$ to 9.2 ppm; m_τ to 61 ppm (parameter-free); the three masses to 0.4 %; and *new* (v2): m_e **computed** to $+1.4/+2.2$ %, zero parameters. All *falsifiable*.

What is solid. $Q = 2/3$ and the m_τ prediction (61 ppm) depend only on $r = \sqrt{2}$ (one number) and the $\mathbb{Z}_3$ structure: parameter-free results that the Standard Model does not reproduce.

What remains cautious. Version 1 estimated at 15–25 % the probability that this structure is fundamental physics rather than a fit — an estimate made *before* the chain of section 6. Version 2 adds: a carrier for $\sqrt{2}$, a derived decomposition of δ, a scale identification at 5×10^{-5}, and an *independent* computation of m_e at ± 3 % with no parameters — four *conditional* corroborations (declared identifications, never a fit). Honesty forbids converting this bundle into certainty: each link can fall with its identification; but the "numerical coincidence" hypothesis must now *also* explain why a Green computation on the ideal trefoil tube recovers m_e to a few percent. It is, by a wide margin, the most constraining and predictive result of the framework.

9 Falsifiability and conclusion

$Q = 2/3$ is *already* confronted with experiment at 9.2 ppm; any future improvement on m_τ or m_μ can **refute** it. The framework further predicts that the same $\mathbb{Z}_3$ symmetry appears as the SU(3) *colour* in the quark sector (where the Koide relation is deformed), sharing the common scale $m_p/3$.

The Standard Model treats the three lepton masses as free parameters and does not explain the Koide relation. TQNT gives it a *structural* reading: three generations = three $\mathbb{Z}_3$ modes of the trefoil, $Q = 2/3$ from $r = \sqrt{2}$ — now *carried* (the two circulation senses, conditional) —, Koide's (1981) relation for m_τ confirmed to 61 ppm, the phase *decomposed* ($\pi/12 - \varepsilon$: the electron's lightness is a nodal protection) and, new in this v2, m_e *computed* to ± 3 % with no parameters — a falsifiable, presently verified result.

References

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